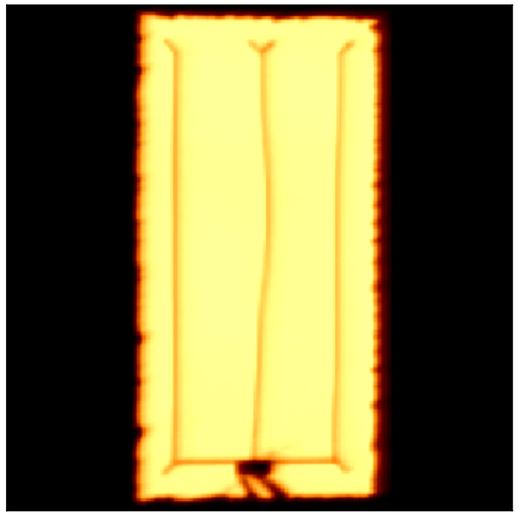
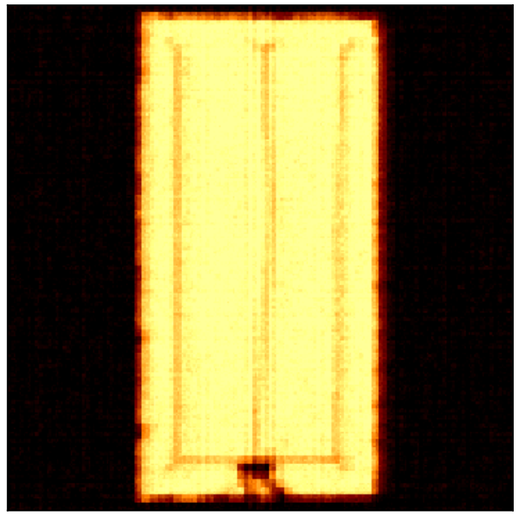
CIGS 0279 | 20% NPL noisy PCM | matched acquisition

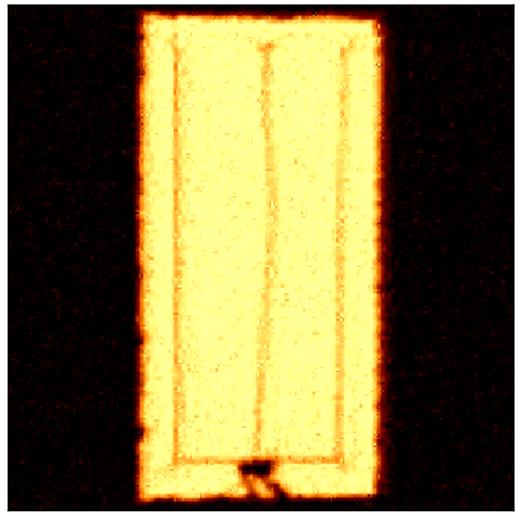
Target
Ground truth



Zero-filled
PSNR 26.51 dB SSIM 0.5039
HaarPSI 0.6453 MSE 2.23e-03



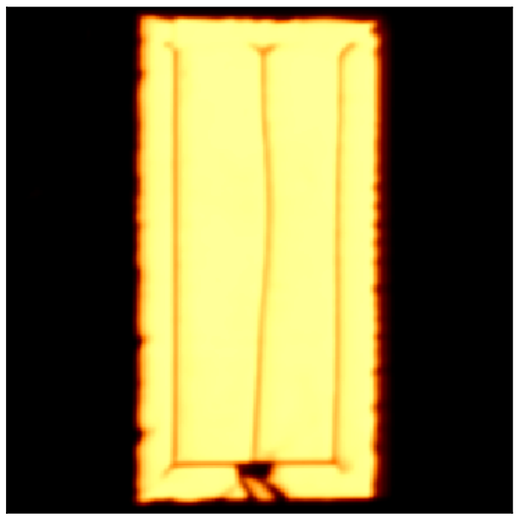
SPGL1-BP
PSNR 27.01 dB SSIM 0.4674
HaarPSI 0.7333 MSE 1.99e-03



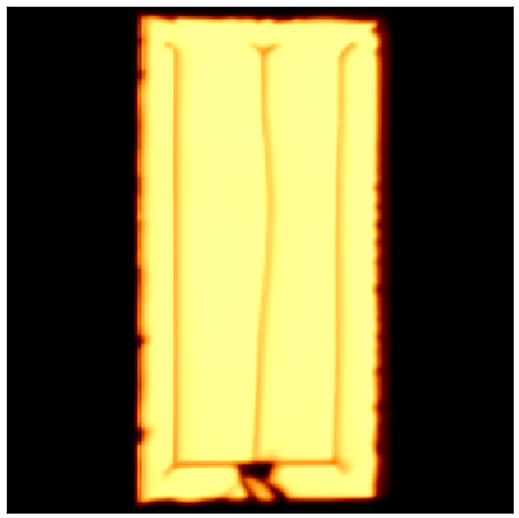
SPGL1-BPDN
PSNR 32.31 dB SSIM 0.8341
HaarPSI 0.7670 MSE 5.88e-04



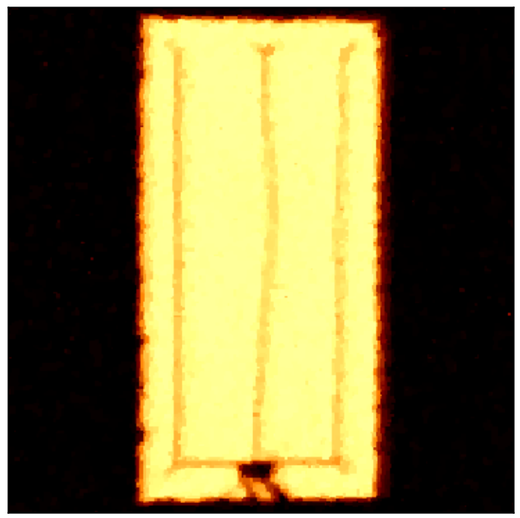
PnP-ADMM oracle
PSNR 40.27 dB SSIM 0.9877
HaarPSI 0.9597 MSE 9.39e-05



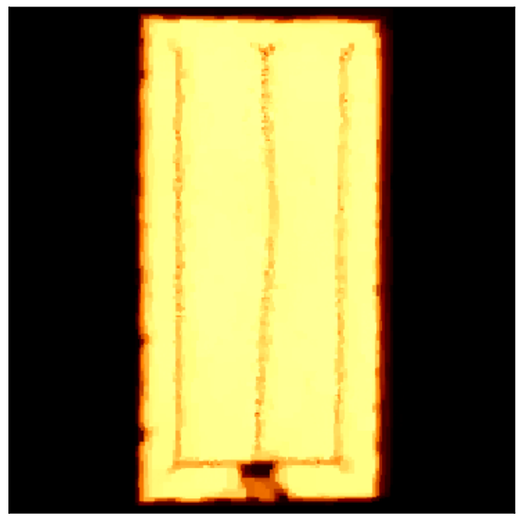
GS-PnP oracle
PSNR 37.63 dB SSIM 0.9838
HaarPSI 0.9586 MSE 1.72e-04



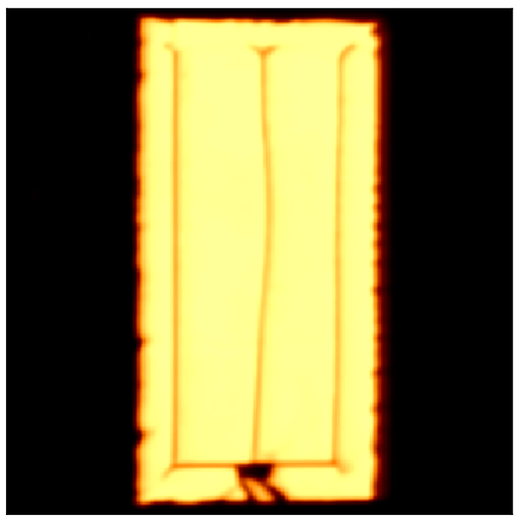
TV
PSNR 31.78 dB SSIM 0.7676
HaarPSI 0.8369 MSE 6.64e-04



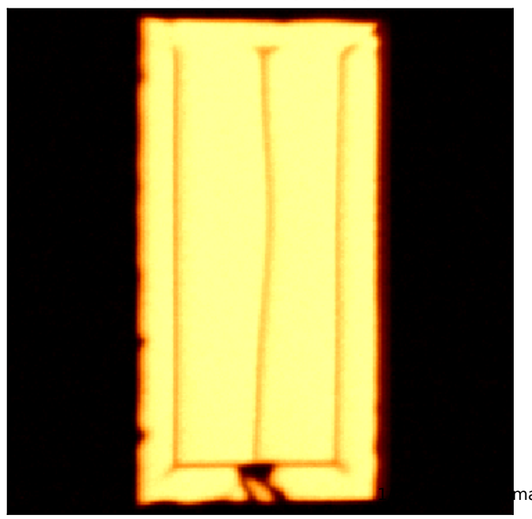
UTV-C60
PSNR 33.42 dB SSIM 0.9372
HaarPSI 0.8287 MSE 4.55e-04



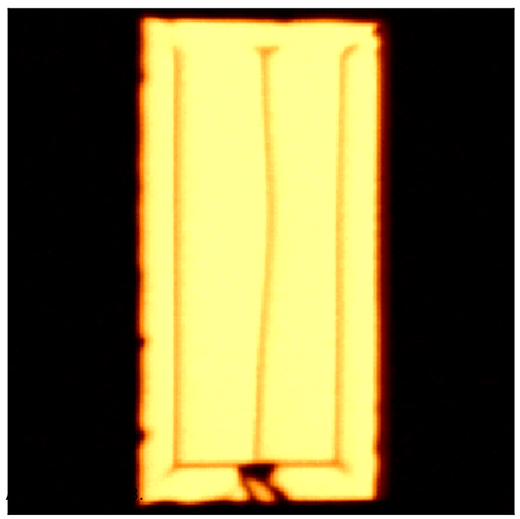
PnP-ADMM tuned-5
PSNR 40.27 dB SSIM 0.9877
HaarPSI 0.9597 MSE 9.39e-05



GS-PnP tuned-5
PSNR 35.70 dB SSIM 0.8774
HaarPSI 0.9121 MSE 2.69e-04



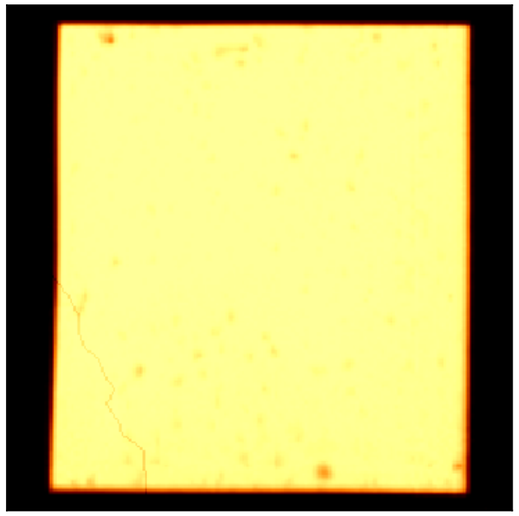
GS-PnP expanded-5
PSNR 35.92 dB SSIM 0.8864
HaarPSI 0.9144 MSE 2.56e-04



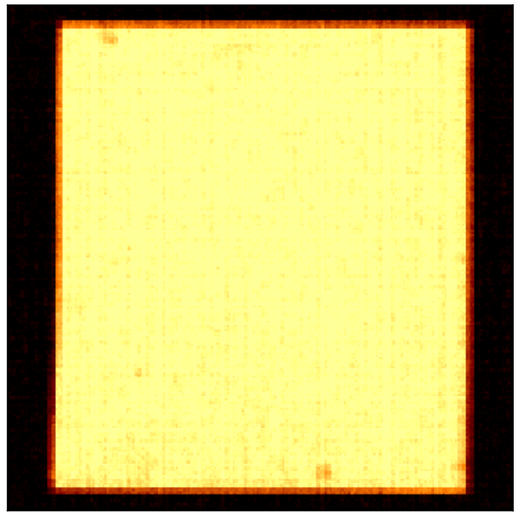
mage display and

Organic 0331 | 20% NPL noisy PCM | matched acquisition

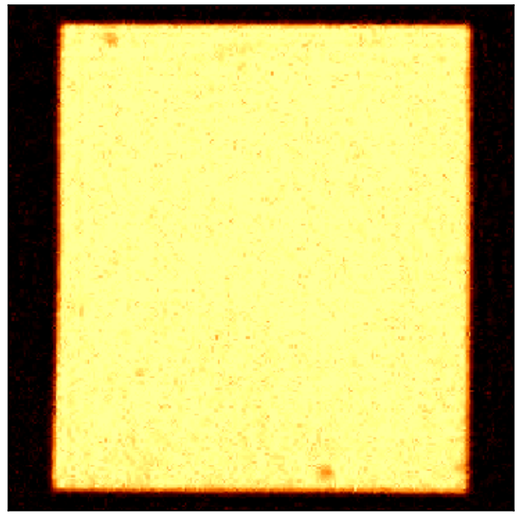
Target
Ground truth



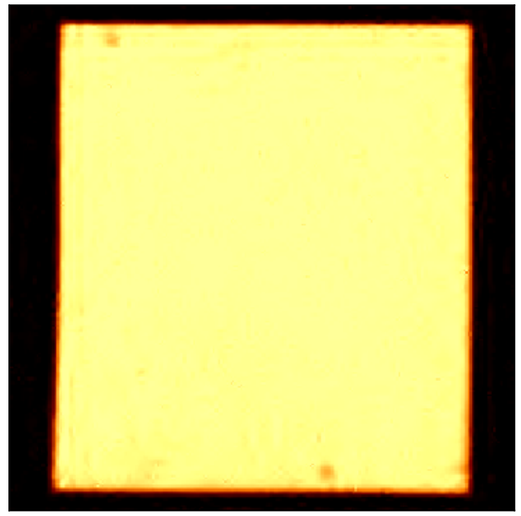
Zero-filled
PSNR 24.46 dB SSIM 0.4570
HaarPSI 0.5363 MSE 3.58e-03



SPGL1-BP
PSNR 26.42 dB SSIM 0.4245
HaarPSI 0.6878 MSE 2.28e-03



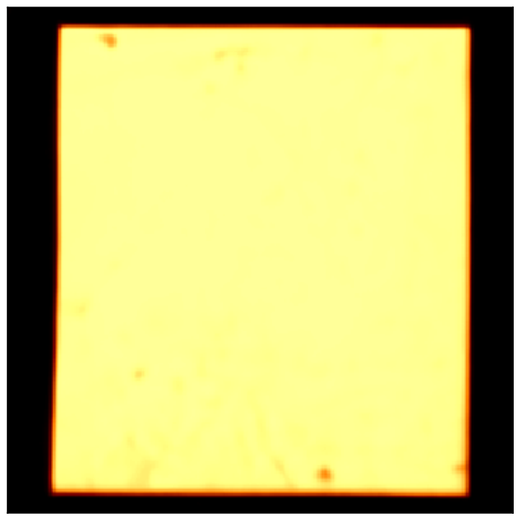
SPGL1-BPDN
PSNR 32.31 dB SSIM 0.8019
HaarPSI 0.7152 MSE 5.88e-04



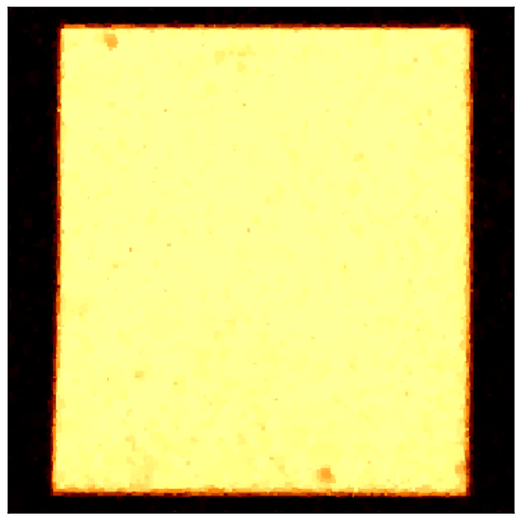
PnP-ADMM oracle
PSNR 38.82 dB SSIM 0.9464
HaarPSI 0.9226 MSE 1.31e-04



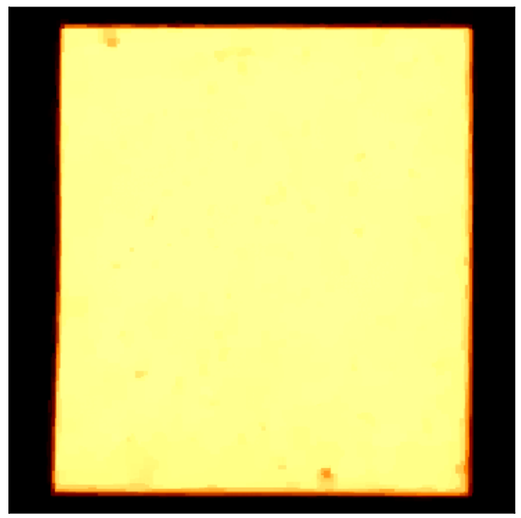
GS-PnP oracle
PSNR 39.39 dB SSIM 0.9512
HaarPSI 0.9252 MSE 1.15e-04



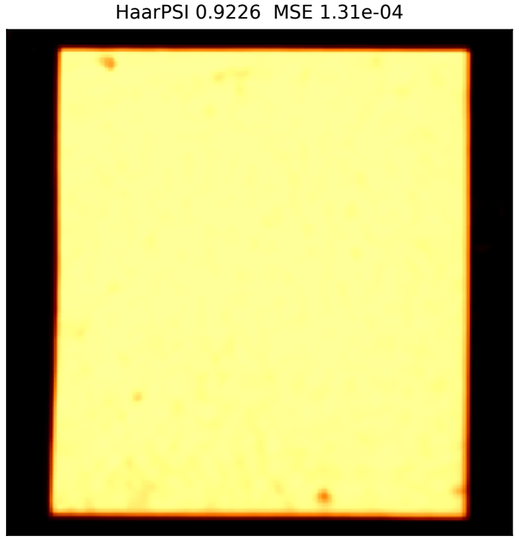
TV
PSNR 31.34 dB SSIM 0.7699
HaarPSI 0.8253 MSE 7.35e-04



UTV-C60
PSNR 36.33 dB SSIM 0.9385
HaarPSI 0.8826 MSE 2.33e-04



PnP-ADMM tuned-5
PSNR 38.82 dB SSIM 0.9464
HaarPSI 0.9226 MSE 1.31e-04



GS-PnP tuned-5
PSNR 34.82 dB SSIM 0.8188
HaarPSI 0.8935 MSE 3.29e-04



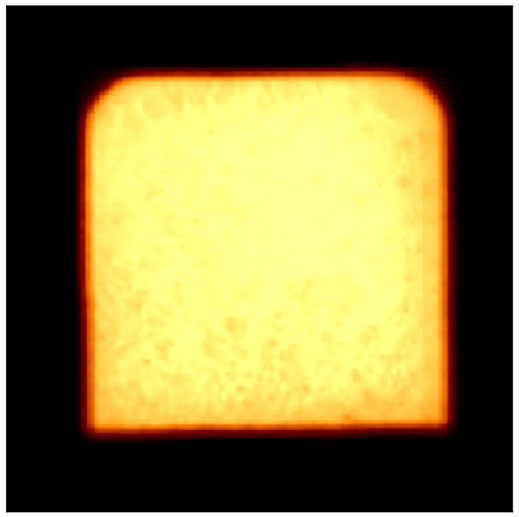
GS-PnP expanded-5
PSNR 35.05 dB SSIM 0.8289
HaarPSI 0.8955 MSE 3.12e-04



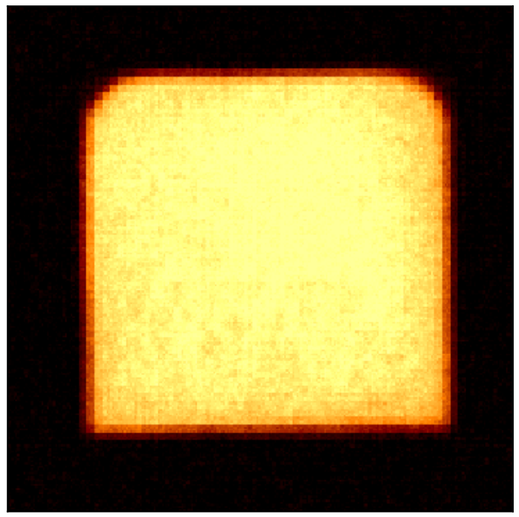
2/5, same l mage display and ot cap 0.8

Perovskite 0786 | 20% NPL noisy PCM | matched acquisition

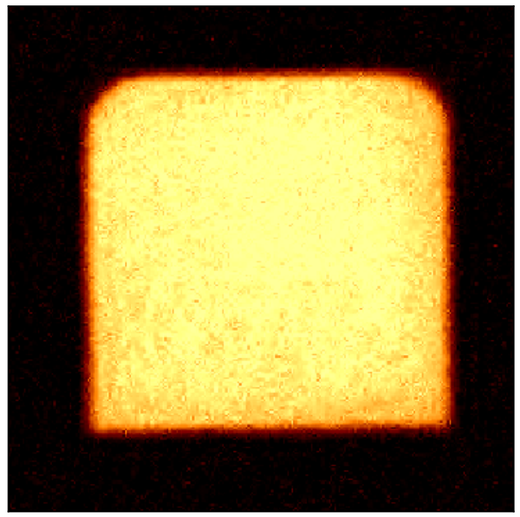
Target
Ground truth



Zero-filled
PSNR 28.55 dB SSIM 0.5481
HaarPSI 0.6027 MSE 1.40e-03



SPGL1-BP
PSNR 27.25 dB SSIM 0.4442
HaarPSI 0.7617 MSE 1.88e-03



SPGL1-BPDN
PSNR 34.84 dB SSIM 0.8168
HaarPSI 0.8200 MSE 3.28e-04



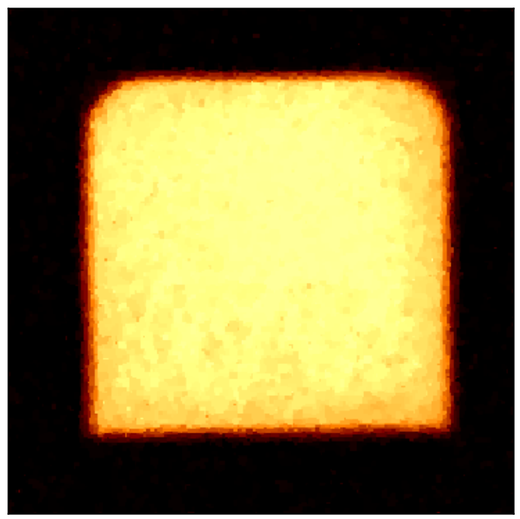
PnP-ADMM oracle
PSNR 38.50 dB SSIM 0.9495
HaarPSI 0.9183 MSE 1.41e-04



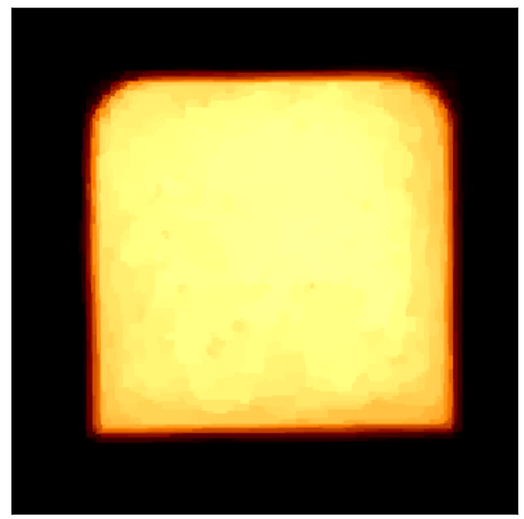
GS-PnP oracle
PSNR 38.37 dB SSIM 0.9409
HaarPSI 0.9149 MSE 1.46e-04



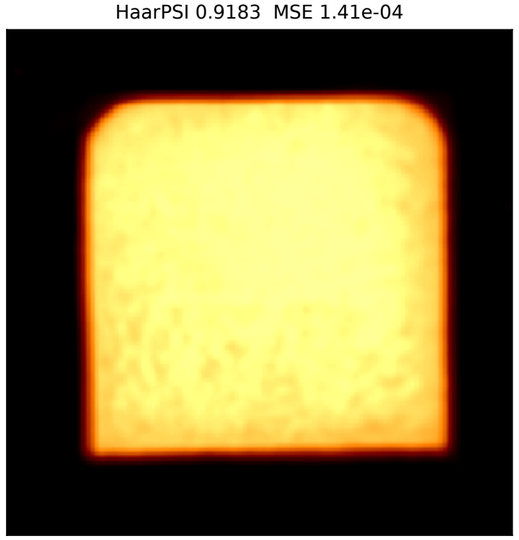
TV
PSNR 32.53 dB SSIM 0.7544
HaarPSI 0.8356 MSE 5.58e-04



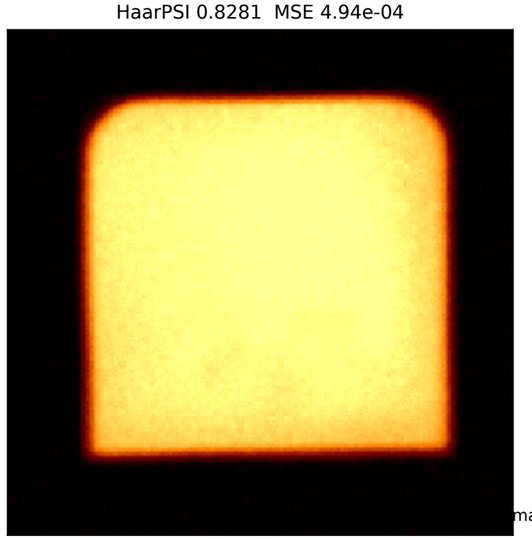
UTV-C60
PSNR 36.41 dB SSIM 0.8960
HaarPSI 0.8587 MSE 2.28e-04



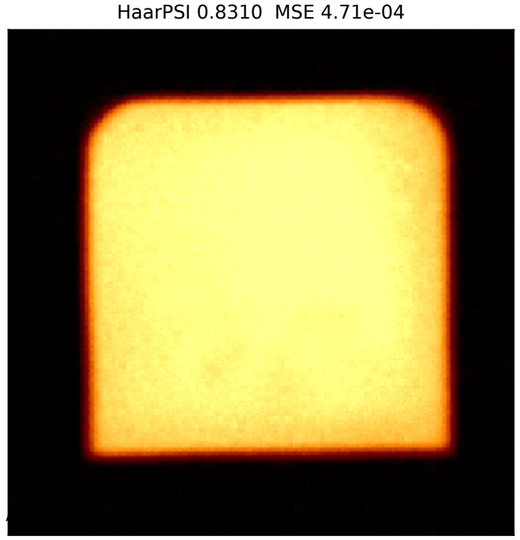
PnP-ADMM tuned-5
PSNR 38.50 dB SSIM 0.9495
HaarPSI 0.9183 MSE 1.41e-04



GS-PnP tuned-5
PSNR 33.06 dB SSIM 0.7270
HaarPSI 0.8281 MSE 4.94e-04

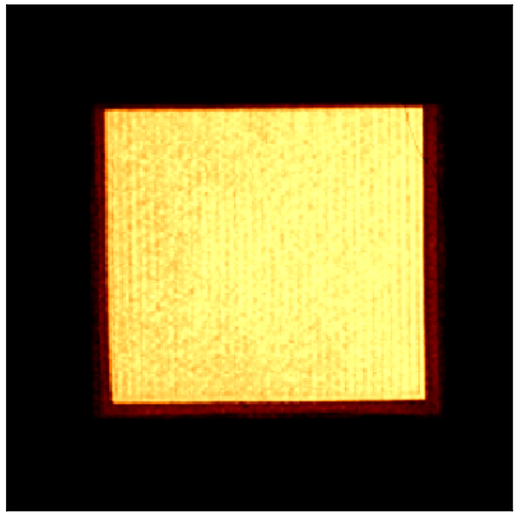


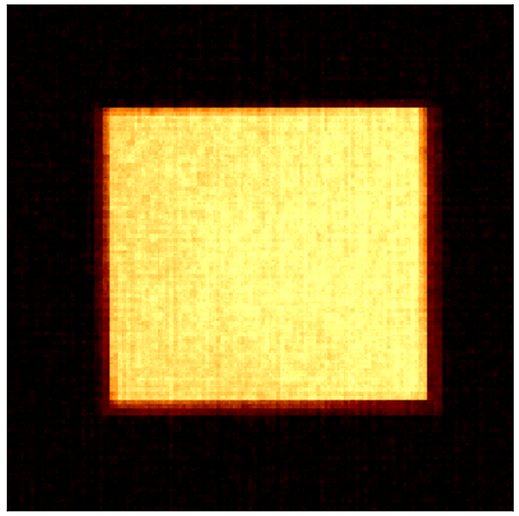
GS-PnP expanded-5
PSNR 33.27 dB SSIM 0.7391
HaarPSI 0.8310 MSE 4.71e-04

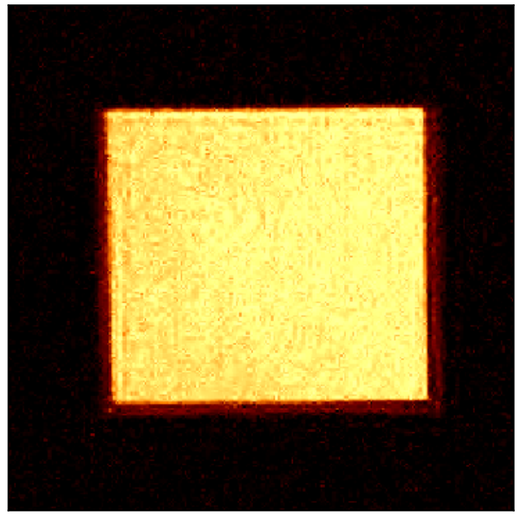


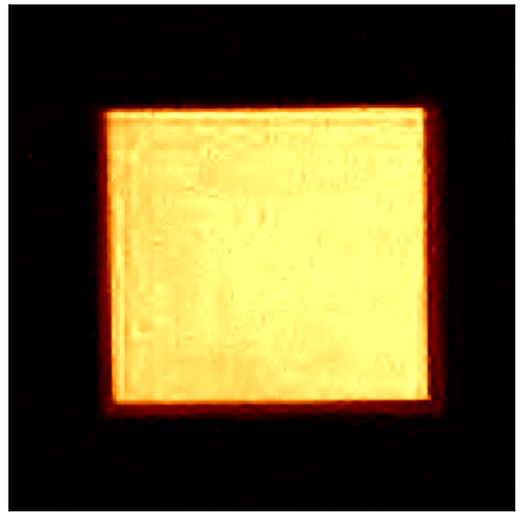
mage display and

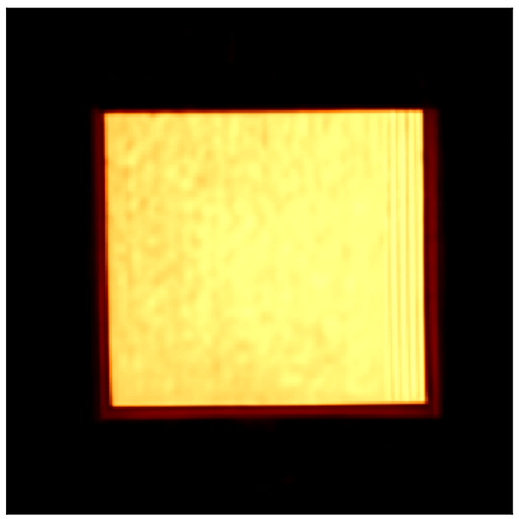
Reference 0279 | 20% NPL noisy PCM | matched acquisition

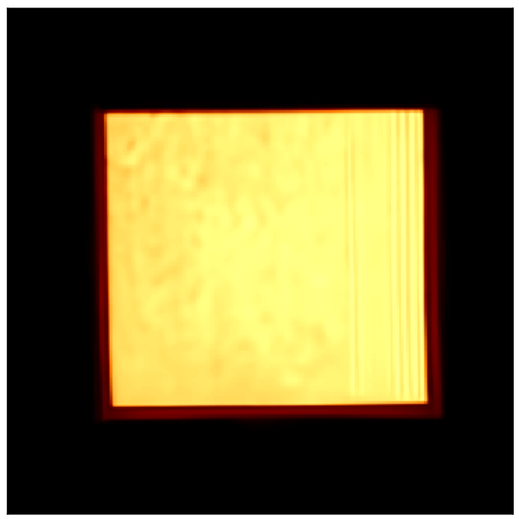

Target
Ground truth

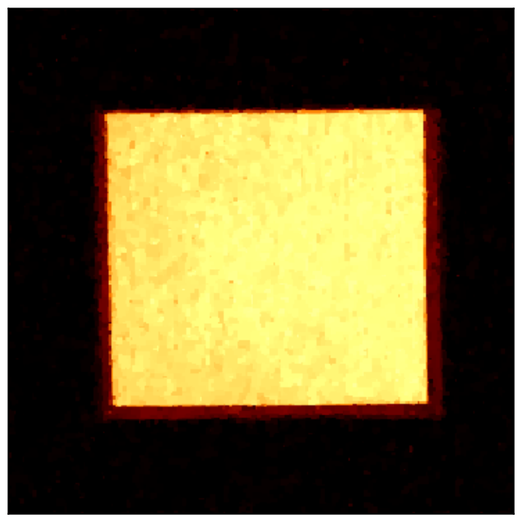

Zero-filled
PSNR 26.06 dB SSIM 0.4455
HaarPSI 0.5445 MSE 2.48e-03

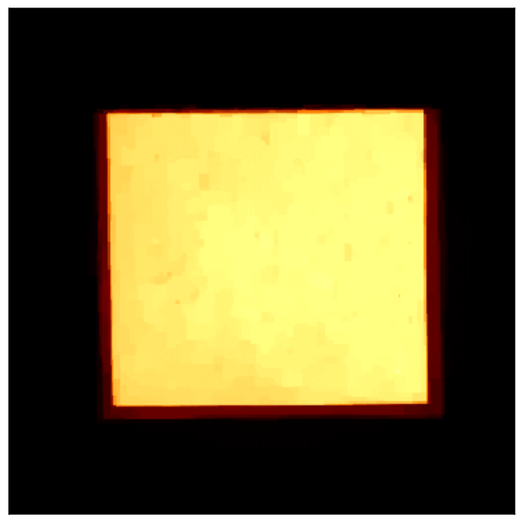

SPGL1-BP
PSNR 26.10 dB SSIM 0.3902
HaarPSI 0.6396 MSE 2.45e-03

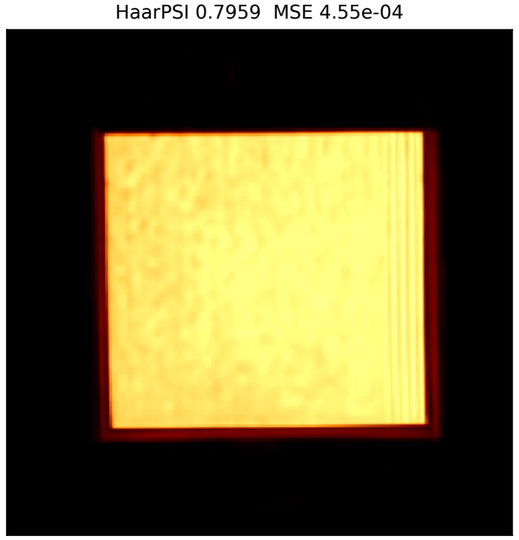

SPGL1-BPDN
PSNR 29.65 dB SSIM 0.6871
HaarPSI 0.6210 MSE 1.08e-03

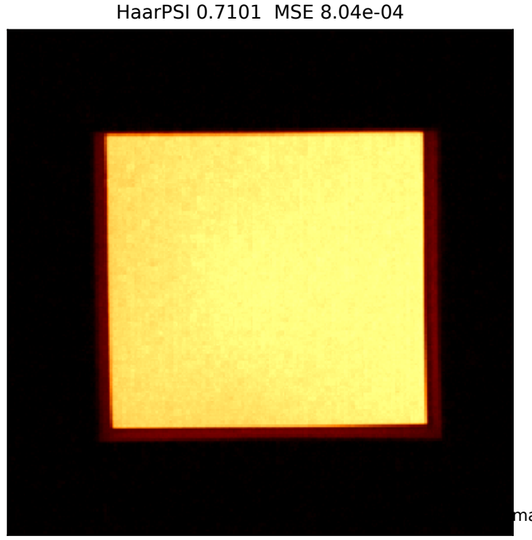

PnP-ADMM oracle
PSNR 33.42 dB SSIM 0.8472
HaarPSI 0.7959 MSE 4.55e-04

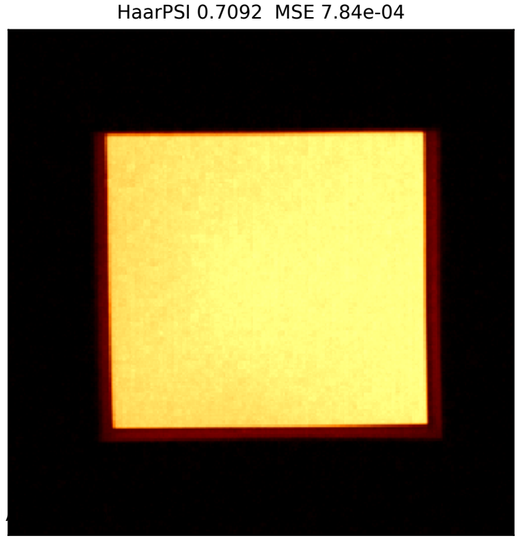

GS-PnP oracle
PSNR 33.37 dB SSIM 0.8367
HaarPSI 0.7776 MSE 4.60e-04


TV
PSNR 30.79 dB SSIM 0.6468
HaarPSI 0.7489 MSE 8.33e-04


UTV-C60
PSNR 32.50 dB SSIM 0.7778
HaarPSI 0.6790 MSE 5.62e-04

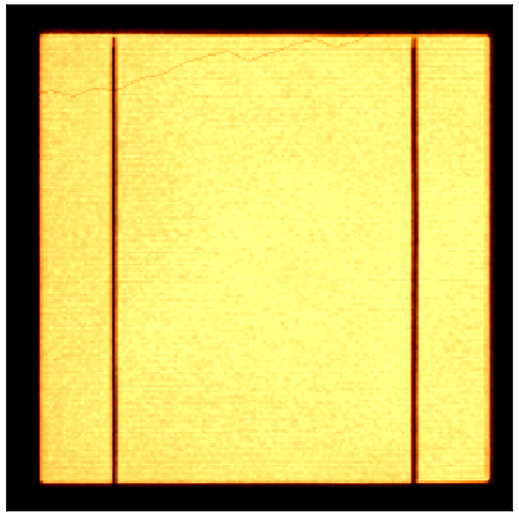

PnP-ADMM tuned-5
PSNR 33.42 dB SSIM 0.8472
HaarPSI 0.7959 MSE 4.55e-04


GS-PnP tuned-5
PSNR 30.95 dB SSIM 0.6659
HaarPSI 0.7101 MSE 8.04e-04

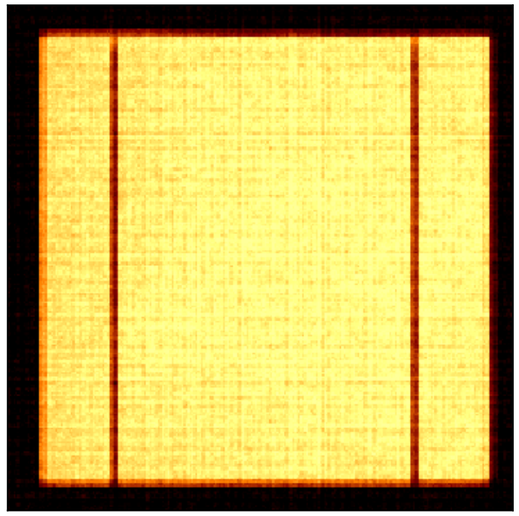

GS-PnP expanded-5
PSNR 31.06 dB SSIM 0.6763
HaarPSI 0.7092 MSE 7.84e-04

Silicone 0370 | 20% NPL noisy PCM | matched acquisition

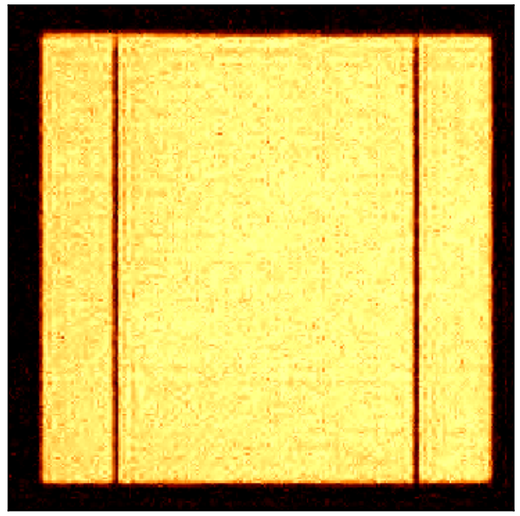
Target
Ground truth



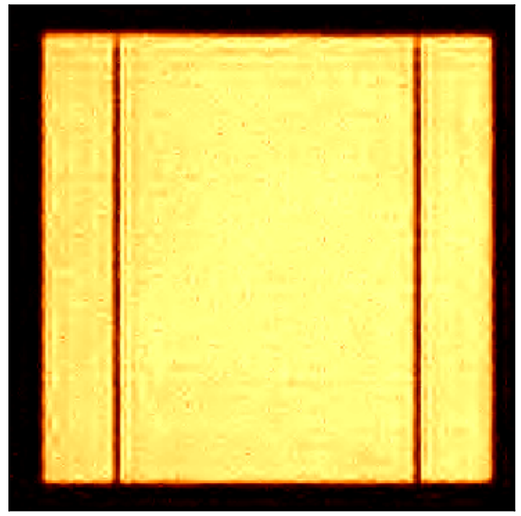
Zero-filled
PSNR 22.77 dB SSIM 0.4300
HaarPSI 0.5323 MSE 5.28e-03



SPGL1-BP
PSNR 24.16 dB SSIM 0.4150
HaarPSI 0.6017 MSE 3.84e-03



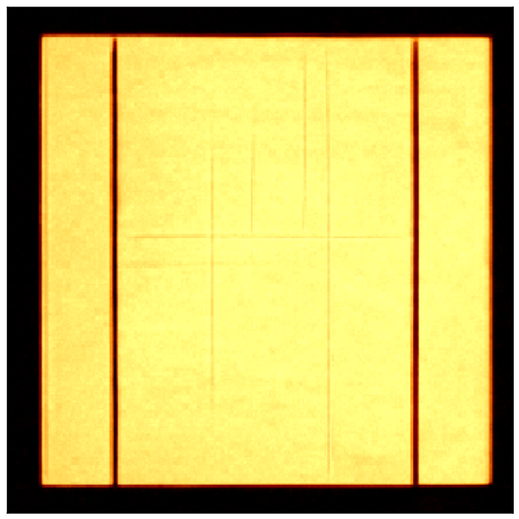
SPGL1-BPDN
PSNR 26.60 dB SSIM 0.6221
HaarPSI 0.5955 MSE 2.19e-03



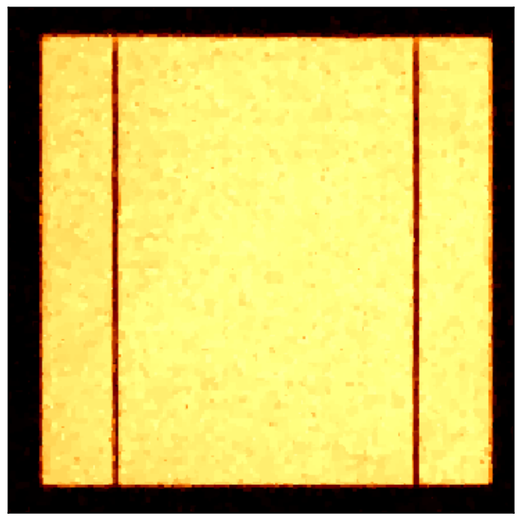
PnP-ADMM oracle
PSNR 32.01 dB SSIM 0.7846
HaarPSI 0.8737 MSE 6.29e-04



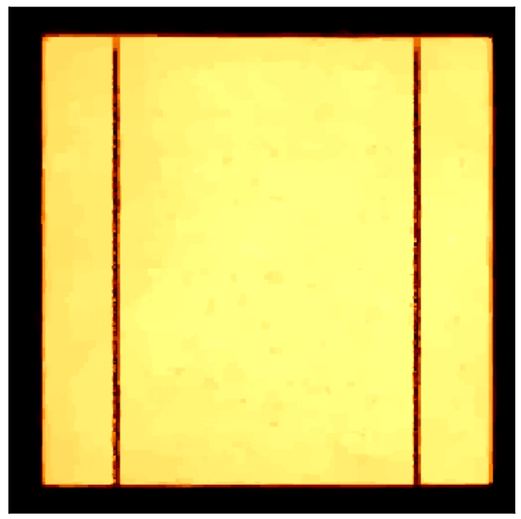
GS-PnP oracle
PSNR 31.14 dB SSIM 0.7265
HaarPSI 0.8640 MSE 7.69e-04



TV
PSNR 28.09 dB SSIM 0.6676
HaarPSI 0.8062 MSE 1.55e-03



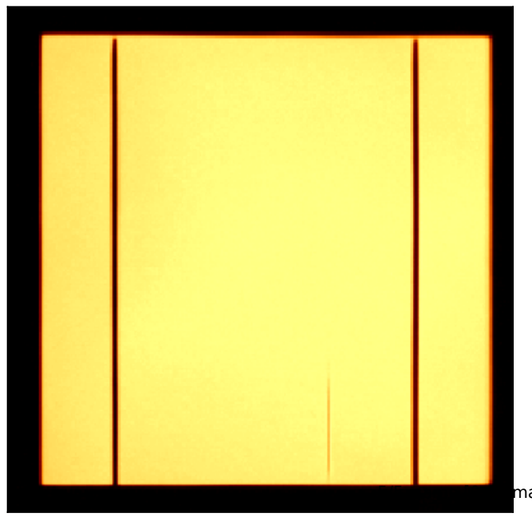
UTV-C60
PSNR 30.60 dB SSIM 0.7764
HaarPSI 0.7768 MSE 8.71e-04



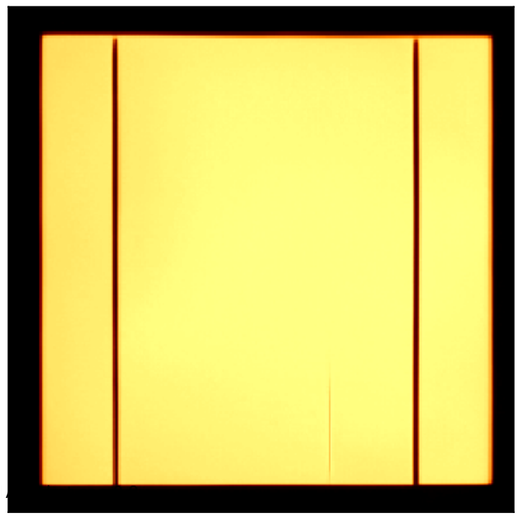
PnP-ADMM tuned-5
PSNR 32.01 dB SSIM 0.7846
HaarPSI 0.8737 MSE 6.29e-04



GS-PnP tuned-5
PSNR 33.19 dB SSIM 0.7819
HaarPSI 0.8494 MSE 4.80e-04



GS-PnP expanded-5
PSNR 33.15 dB SSIM 0.7808
HaarPSI 0.8356 MSE 4.84e-04



mage display and A